GOVERNMENT OF INDIA • COPYRIGHT OFFICE

DEPARTMENT FOR PROMOTION OF INDUSTRY AND INTERNAL TRADE (DPIIT)

APPLICATION FOR COPYRIGHT REGISTRATION — FORM XIV

DOCUMENT 01

SOFTWARE PROJECT DESCRIPTION

Comprehensive Technical & Statutory Specification of the Frontend Software Submitted in Support of Copyright Registration (Form XIV)

Statutory Category:	Computer Software / Literary Work (Section 2(o), Copyright Act, 1957)
Software Title:	ARB SoftTech AI Enterprise Voice Calling Agent Platform — Frontend Software
Application Package:	voxora-enterprise-ai-voice (Version 2.0.0 Production)
Languages & Tech:	TypeScript (v5.6.3), JavaScript (ES2022), React 18.3.1, Vite 6, Tailwind CSS
Target Authority:	Registrar of Copyrights, Copyright Office, New Delhi, India
Statutory Mandate:	Deposit under Rule 70(5), Copyright Rules, 2013 (First 10 & Last 10 Pages Format)
Date of Compilation:	October 2026

OFFICIAL NOTICE OF COPYRIGHT DEPOSIT

CONFIDENTIAL STATUTORY SUBMISSION: This official documentation package is prepared strictly for submission to the Registrar of Copyrights, Government of India, under Section 45 of the Copyright Act, 1957. All proprietary source code, user interface designs, and architecture representations herein reflect original works of authorship. Third-party open-source libraries are disclaimed under Section 52.

1. STATUTORY SOFTWARE IDENTIFICATION & CLASSIFICATION

This project description is formally submitted pursuant to Rule 70 of the Copyright Rules, 2013 accompanying Form XIV (Application for Registration of Copyright) before the Registrar of Copyrights, Department for Promotion of Industry and Internal Trade (DPIIT), New Delhi, India. The subject matter of registration is an original computer software work comprising the interactive frontend client application for enterprise telephony voice operations.

Statutory Parameter	Particulars / Formal Record
Title of the Work	ARB SoftTech AI Enterprise Voice Calling Agent Platform — Frontend Software
Statutory Category	Literary Work — Computer Software under Section 2(o) & Section 2(ffc) of the Copyright Act, 1957
Package Identifier	voxora-enterprise-ai-voice (Version 2.0.0 Production)
Primary Languages	TypeScript (v5.6.3), JavaScript (ES2022 / ECMAScript Modules), HTML5, CSS3
Core Architecture	Client-side Single Page Application (SPA) built on React 18.3.1 and Vite 6.0.5
Codebase Scope	118 original source files | 23,552 lines of code | 941 KB source size (excluding node_modules)
Jurisdiction & Location	Republic of India

2. PURPOSE OF THE SOFTWARE & EXECUTIVE OVERVIEW

The **ARB SoftTech AI Enterprise Voice Calling Agent Platform (Frontend Software)** is an enterprise-grade web-based operational console designed to configure, monitor, dispatch, and analyze autonomous AI-driven PSTN telephone calls. The platform bridges complex telecommunications carrier trunks (VoBiz, Twilio) and conversational artificial intelligence models (Sarvam AI for multilingual Indic speech and Google Gemini for post-call lead qualification) into an intuitive, high-performance visual dashboard.

- **Real-Time Telephonic Telemetry:** Sub-second polling and WebSocket state visualization for concurrent ongoing mobile/landline telephone calls across Indian and global PSTN networks.
- **Conversational Voice Agent Workspace:** Visual prompt engineering workbench allowing non-technical operators to create custom voice personas, tune speech temperature, set interruptibility delays, and assign Hindi (Devanagari) or Indic acoustic models.
- **Lead Intelligence Pipeline:** Automated visualization of post-call conversation analytics, extracting customer names, phone numbers, locations, budgets, and callback times, accompanied by hoverable AI reasoning chips.
- **Interactive Branching Questionnaire Builder:** Dynamic flowchart editor allowing business users to define multi-turn conversational question logic with bulk Excel (.xlsx) and Word (.docx) import capabilities.
- **Dual Look Visual Architecture:** A proprietary design system permitting seamless real-time switching between a corporate 'Classic Enterprise' layout and a futuristic 'Modern High-Tech' dark glassmorphic cockpit.

3. MAJOR FUNCTIONAL MODULES & SUBSYSTEMS

Module 1: Dashboard & Live Telemetry

Source Path: src/features/dashboard/, src/components/ui/MetricCard.tsx

Presents high-level operational metrics including aggregate call volume, connection rates, total minutes consumed, and lead conversion rates. Features animated Recharts graphics, active agent popovers, and quick-dispatch call buttons.

Module 2: AI Voice Agent Studio

Source Path: src/features/agents/, AgentFormDrawer.tsx, AgentWorkspacePage.tsx

A comprehensive workspace for authoring voice agents. Features multilingual TTS persona selectors (Sarvam AI Shubh, Ritu, Aditya), LLM prompt templates, barge-in sensitivity sliders, temperature tuning, and one-click PDF dossier export.

Module 3: Active Ongoing Calls Monitor

Source Path: src/features/calls/ActiveCallsPage.tsx, QueuePage.tsx

Real-time visual console displaying active call channels, connection durations, caller phone numbers, and live audio waveform animations.

Module 4: Historical Call Detail Records & Drawer

Source Path: src/features/calls/CallsPage.tsx, CallDetailDrawer.tsx

Full CDR audit table with text search, date filtering, and client-side Excel export. Opening any call launches a lateral drawer with turn-by-turn speaker-diarized transcripts, AI interest reasoning, and embedded 8kHz WAV audio player.

Module 5: Lead Intelligence & Intent Pipeline

Source Path: src/features/leads/LeadsPage.tsx

Interactive sales qualification kanban/table displaying customer profiles extracted from transcripts (Name, Email, City, Budget, Preferred Callback Time). Features why-interested reason chips with hoverable evidence tooltips.

Module 6: Conversational Questionnaire Engine

Source Path: src/features/questionnaires/, QuestionnaireEditorPage.tsx

Interactive visual flow editor for branching conversational trees. Includes specialized modal parsers for bulk question import from Microsoft Excel (.xlsx) and Microsoft Word (.docx) files.

Module 7: Outbound PSTN Quick-Dial Modal

Source Path: src/components/layout/QuickDialModal.tsx, src/utils/quickDial.ts

Global dialer modal allowing instant outbound phone call initiation to any mobile/landline number via backend carrier proxy dispatch.

Module 8: Telephony Carrier & AI Settings

Source Path: src/features/settings/, SarvamSettings.tsx, VoBizTelephonySettings.tsx

Secure credential management interfaces for Sarvam AI speech models, VoBiz carrier trunks, Twilio SIP trunks, and Google Gemini transcript intelligence APIs.

Module 9: Model Lab & Latency Diagnostic Suite

Source Path: src/features/model_lab/ModelLabPage.tsx

Specialized testing workbench allowing live voice model benchmarking, STT/TTS latency profiling, and telephony audio stream inspection.

Module 10: Multi-Tenant Business Administration

Source Path: src/features/businesses/, BusinessProvider.tsx, BusinessSelector.tsx

Multi-tenant workspace manager allowing operators to isolate telephony trunks, agents, and lead records by client entity, with PDF profile export.

Module 11: Follow-ups, Contacts & Invoicing

Source Path: src/features/followups/, src/features/contacts/, src/features/invoices/

CRM contact book with DNC (Do-Not-Call) flags, scheduled callback task manager, and minute consumption invoicing module.

Module 12: Dual Look & Modern Visual Design System

Source Path: src/features/modern/, LookToggle.tsx, GlowProvider.tsx, ParticleStars.tsx

Advanced styling engine providing 17 dedicated modern views, ambient radial gradients, interactive cursor glow effects, and HTML5 canvas particle stars.

4. TECHNOLOGY STACK & FRONTEND ARCHITECTURE

Architectural Layer	Technologies & Version	Architectural Role & Justification
Language Runtime	TypeScript 5.6.3 JavaScript (ES2022)	Strongly-typed application logic, interface contracts, and compile-time AST validation.
UI Framework	React 18.3.1 (Virtual DOM)	Declarative component hierarchy, concurrent rendering, and hook-based lifecycle.
Build & Bundler	Vite 6.0.5 + Rollup	High-speed development server with HMR and optimized production bundle compilation.
Routing Engine	React Router DOM 6.28.1	Client-side declarative routing, URL state synchronization, and dual-look page wrapping.
Server State	TanStack React Query 5.62.7	Asynchronous caching, background telephony polling, and query invalidation.
Client State	Zustand 5.0.2 + React Context	Global theme toggles, modal dialog triggers, and multi-tenant business context.
Design System	Tailwind CSS 3.4.17 + PostCSS	Utility-first design tokens, custom dark-mode CSS variables, and glassmorphic styling.
Data Visualization	Recharts 2.15.0 + Lucide React	Responsive SVG telephony graphs, call duration charts, and vector iconography.
Document Exports	jsPDF 4.2.1 + SheetJS (xlsx)	Client-side dynamic generation of PDF agent dossiers and XLSX lead reports.
Deployment	Docker + Nginx Alpine	Containerized production distribution with Nginx reverse proxy and SPA routing fallback.

5. DEVELOPMENT METHODOLOGY & AI-ASSISTED DISCLOSURE

In strict adherence to the principles of transparency and statutory compliance under the Indian Copyright Act, 1957, this section details the development methodology and the utilization of artificial intelligence tools during the engineering of the frontend client application.

AI-ASSISTED SOFTWARE AUTHORSHIP DISCLOSURE

STATUTORY DISCLOSURE ON GOOGLE ANTIGRAVITY USAGE:
The frontend software was authored using **Google Antigravity**, an advanced agentic AI-assisted software engineering development environment.

1. Status under Indian Copyright Law: Under Section 2(d)(i) of the Indian Copyright Act, 1957, an author of a computer programme must be a natural human person. Google Antigravity is not claimed or identified as an author or co-author. It served strictly as a computational instrument and intelligent programming aid.

2. Human Intellectual Effort & Originality: Human software engineers conceived the product requirements, formulated architectural blueprints, designed user journeys, defined domain data models in api.ts, crafted customized natural language prompts, debugged integration failures, curated code outputs, and performed comprehensive end-to-end integration and verification. The creative selection, coordination, and arrangement of code elements reflect genuine human intellectual authorship protected under Section 13(1)(a).

6. THIRD-PARTY DEPENDENCIES & OPEN-SOURCE CARVE-OUTS

The application integrates standardized open-source libraries (e.g. React, React Router, Vite, Axios, Tailwind CSS, Recharts) distributed under permissive open-source licenses (MIT, Apache 2.0). In compliance with Section 52 of the Indian Copyright Act, 1957 and official filing guidelines:
• The Applicant **expressly disclaims** proprietary copyright over all third-party libraries.
• Third-party code is excluded from the proprietary claim.
• A complete inventory of all 19 runtime dependencies and 10 build tooling dependencies is recorded in THIRD_PARTY_DEPENDENCIES.md.

7. SCOPE OF COPYRIGHT CLAIMED & LEGAL LIMITATIONS

Copyright protection under Section 14(b) of the Indian Copyright Act, 1957 is claimed strictly over:
1. **Original Source Code:** The original TypeScript, React, and CSS source code files authored for this platform.
2. **User Interface Component Architecture:** The proprietary structural hierarchy of interactive components, modals, and drawers.

3. **Telephony Integration Logic:** Custom state machines and REST/WebSocket integration routines managing PSTN voice operations.
4. **Dynamic Visual Styling:** Custom ambient CSS animations, glow effects, and modern dual-look presentation layers.
5. **Selection and Arrangement:** The creative coordination and compilation of modules into a unified enterprise voice calling platform.

8. FORM XIV STATEMENT OF PARTICULARS REFERENCE MAPPING

Form XIV Column	Statutory Query	Formal Description / Document Reference
Col 2	Name & Address of Applicant	Refer to Section 2 of 05_Development_Ownership_Declaration.pdf.
Col 4	Class & Description of Work	Literary (Computer Software). Single Page Application for enterprise voice operations.
Col 5	Title of Work	ARB SoftTech AI Enterprise Voice Calling Agent Platform — Frontend Software
Col 6	Language of Work	TypeScript (v5.6.3), JavaScript (ES2022), HTML5, CSS3
Col 7	Name & Nationality of Author	Natural human software engineer(s). Refer to Document 05.
Col 8 & 9	Publication Status	Unpublished proprietary software / Completed October 2026 in India.
SoFP Col 1	Originality of Work	Yes, an original computer software literary work authored for the applicant.
SoFP Col 3	Third-Party Materials	Permissive open-source dependencies disclaimed under Section 52.

Prepared by:

_____
Senior Software Documentation Engineer
Date: 31st October 2026

Verified on behalf of Applicant:

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Authorized Officer / Signatory
For: ARB SoftTech